MATH 111 Calculus 1

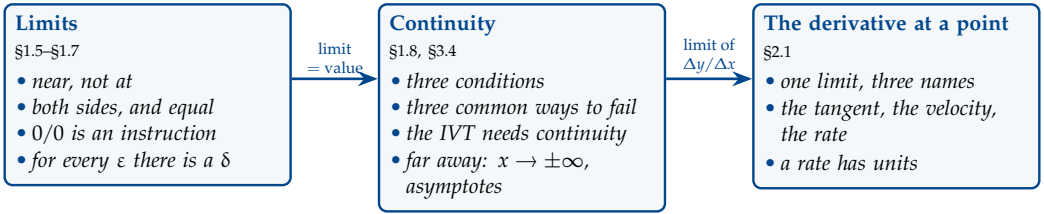
Revision

Limits, continuity and the derivative at a point — in questions

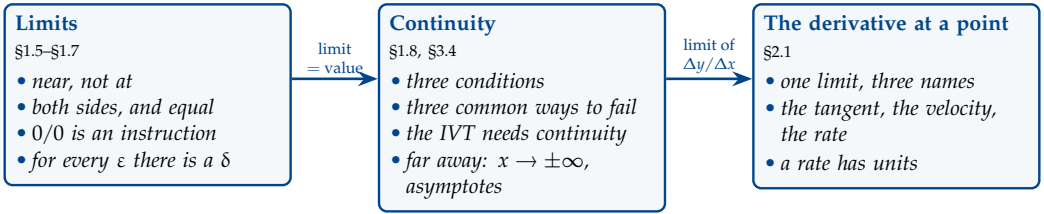
Week 7 · Lecture 17 · §1.5–§1.8, §3.4, §2.1

Dr. Salnikova Chekeres · New Uzbekistan University · Fall 2026

The map — limits, continuity, the derivative at a point



The map — limits, continuity, the derivative at a point



Today

The three boxes are one idea, the limit: continuity asks that the limit *equal* the value, and the derivative at a point *is* a limit — of difference quotients.

STAND UP

"If $\lim_{x \to a} f(x) = 0$, then $\lim_{x \to a} f(x)\, g(x) = 0$, whatever the function g."

Stand up if you think this is TRUE.

Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

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Verdict

FALSE. $f(x) = x$, $g(x) = \dfrac{1}{x}$, $a = 0$: $f(x)\,g(x) = 1$ for every $x \neq 0$, so $\lim_{x\to 0} f(x)\,g(x) = 1$. "Zero times anything" is not a limit law: $0 \cdot \infty$ is an instruction, like $0/0$. What is true: if g is *bounded* near a, $|g(x)| \le M$, then $-M|f(x)| \le f(x)\,g(x) \le M|f(x)|$, and the Squeeze Theorem gives 0 — that is why $\lim_{x\to 0} x \sin\frac{1}{x} = 0$.

Question 45

$$\lim_{x \to -3} \frac{x^2 - 2x - 15}{x^2 + 4x + 3} = ?$$

A. 4

B. Does not exist — $\frac{0}{0}$.

C. 0 — the numerator is 0 at $x = -3$.

D. ∞ — the denominator is 0 at $x = -3$.

In your team

Write your own letter first — then talk. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

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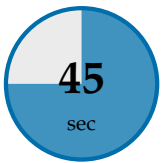


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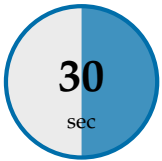
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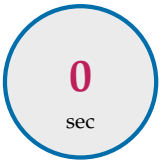
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Hands down. Thank you.

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ANSWER

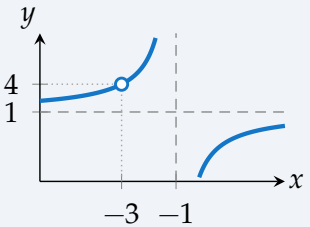
Answer

A. For $x \neq -3$: $\frac{x^2-2x-15}{x^2+4x+3} = \frac{(x+3)(x-5)}{(x+3)(x+1)} = \frac{x-5}{x+1}$, and the limit looks only at such x:
$\lim_{x\to-3} \frac{x^2-2x-15}{x^2+4x+3} = \lim_{x\to-3} \frac{x-5}{x+1} = \frac{-8}{-2} = 4$.

B stops where the work starts.

C forgets that the denominator is 0 too.

D: a zero denominator says where to *look* — here the factor cancels: a hole, not an asymptote.



a hole at $(-3, 4)$

STAND UP

"If f is continuous at a and $f(a) > 0$, then $f(x) > 0$ for every x in some interval $(a - \delta, a + \delta)$."

Stand up if you think this is TRUE.

Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

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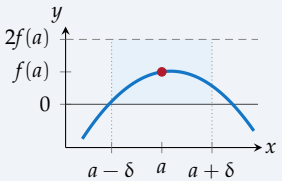
"If f is continuous at a and $f(a) > 0$, then $f(x) > 0$ for every x in some interval $(a - \delta, a + \delta)$."

Stand up if you think this is TRUE.

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Verdict

TRUE. Continuity at a: $\lim_{x \to a} f(x) = f(a)$. Take $\varepsilon = f(a)$ in the definition: there is a $\delta > 0$ with $|f(x) - f(a)| < f(a)$ whenever $|x - a| < \delta$ — and then $f(x) > f(a) - f(a) = 0$. A positive value stays positive nearby. Without continuity it fails: $f(a) = 1$ and $f(x) = -1$ for every $x \neq a$.



on $(a - \delta, a + \delta)$ the graph stays between 0 and $2f(a)$

Question 46

Let $g(x) = \begin{cases} x^2 + k, & x \leq -3, \\ 25 + kx, & x > -3. \end{cases}$ For which k is g continuous everywhere?

A. $k = \dfrac{17}{2}$

B. $k = 4$

C. $k = -8$

D. Every k: both pieces are polynomials, and polynomials are continuous.

In your team

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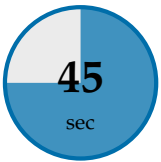
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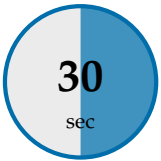
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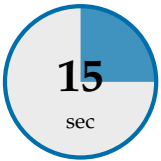
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Hands down. Thank you.

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ANSWER

Answer

B. Away from -3 each piece is a polynomial. At the seam:
$\lim_{x\to -3^-} g(x) = \lim_{x\to -3^-} (x^2 + k) = 9 + k = g(-3)$ and
$\lim_{x\to -3^+} g(x) = \lim_{x\to -3^+} (25 + kx) = 25 - 3k$. Continuity needs $9 + k = 25 - 3k$:
$k = 4$ (both sides 13).

A squares -3 as -9.

C puts $x = 3$, not -3, into $25 + kx$.

D forgets the seam: the pieces must meet.

Question 47

f is continuous on $[2, 8]$, and $f(2) = 2$, $f(5) = -6$, $f(8) = 9$. What does the Intermediate Value Theorem guarantee about the equation $f(x) = 5$?

A. Two solutions: one in $(2, 5)$ and one in $(5, 8)$.

B. Exactly one solution, and it lies in $(5, 8)$.

C. Nothing: the theorem is about zeros, and $f(2)$ and $f(8)$ are both positive.

D. At least one solution in $(5, 8)$.

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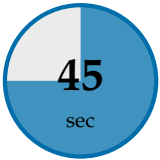
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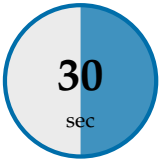
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B. Exactly one solution, and it lies in $(5, 8)$.

C. Nothing: the theorem is about zeros, and $f(2)$ and $f(8)$ are both positive.

D. At least one solution in $(5, 8)$.

Hands down. Thank you.

A B C D

ANSWER

Answer

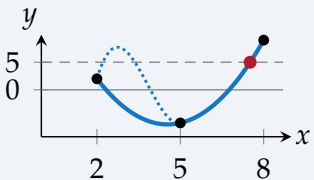
D. On $[5,8]$: $f(5) = -6 < 5 < 9 = f(8)$, so $f(c) = 5$ for some c in $(5,8)$. On $[2,5]$ both values lie below 5: the theorem says nothing there.

A: the hypothesis fails on $[2,5]$.

B: the theorem never counts.

C: any N between the values, not only 0.

A guarantee, not a fence.



solid: once; dotted: 3 times

Question 48

True or false ?

If f is continuous on $\mathbb{R}$ and $f(x)^2 = 1$ for every x, then f is constant.

In your team

Decide your own answer first — true or false — then talk. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 40 seconds

True or false ?

If f is continuous on $\mathbb{R}$ and $f(x)^2 = 1$ for every x, then f is constant.

40
sec

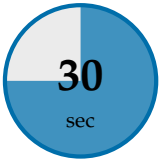
In your team

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THINK & TALK · 40 seconds

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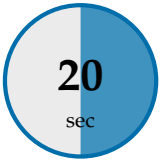
In your team

Decide your own answer first — true or false — then talk. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 40 seconds

True or false ?

If f is continuous on $\mathbb{R}$ and $f(x)^2 = 1$ for every x, then f is constant.



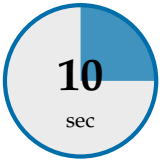
In your team

Decide your own answer first — true or false — then talk. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 40 seconds

True or false ?

If f is continuous on $\mathbb{R}$ and $f(x)^2 = 1$ for every x, then f is constant.



In your team

Decide your own answer first — true or false — then talk. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 40 seconds

True or false ?

If f is continuous on $\mathbb{R}$ and $f(x)^2 = 1$ for every x, then f is constant.

0
sec

In your team

Decide your own answer first — true or false — then talk. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

VOTE

True or false ?

If f is continuous on $\mathbb{R}$ and $f(x)^2 = 1$ for every x, then f is constant.

Eyes on the screen. Hands up when your answer appears — hold for three seconds.

TRUE FALSE

VOTE

True or false ?

If f is continuous on $\mathbb{R}$ and $f(x)^2 = 1$ for every x, then f is constant.

Eyes on the screen. Hands up when your answer appears — hold for three seconds.

TRUE

FALSE

VOTE

True or false ?

If f is continuous on $\mathbb{R}$ and $f(x)^2 = 1$ for every x, then f is constant.

Eyes on the screen. Hands up when your answer appears — hold for three seconds.

TRUE

FALSE

VOTE

True or false ?

If f is continuous on $\mathbb{R}$ and $f(x)^2 = 1$ for every x, then f is constant.

Hands down. Thank you.

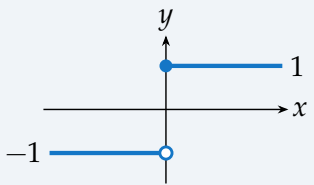
TRUE FALSE

ANSWER

Answer

True. **PROOF** Every value of f is 1 or -1. Suppose $f(a) = 1$ and $f(b) = -1$. f is continuous on the closed interval between a and b, and 0 lies between -1 and 1: by the Intermediate Value Theorem $f(c) = 0$ for some c — but $f(c)^2 = 1$, a contradiction. So f never changes value: $f = 1$ everywhere, or $f = -1$ everywhere. ■

Without continuity it fails: $f(x) = 1$ for $x \geq 0$, $f(x) = -1$ for $x < 0$.



$f^2 = 1$, not continuous, not constant

Question 49

Prove from the definition that $\lim_{x \to 6} (x^2 - 9x) = -18$.

The proof begins: *"Let* $\varepsilon > 0$. *Choose* $\delta = \ldots$*"* Which δ works for *every* $\varepsilon > 0$?

A. $\delta = \dfrac{\varepsilon}{4}$

B. $\delta = \min\left\{1, \dfrac{\varepsilon}{4}\right\}$

C. $\delta = \dfrac{\varepsilon}{|x-3|}$

D. $\delta = \min\left\{1, \dfrac{\varepsilon}{3}\right\}$

In your team

Write your own letter first — then talk. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

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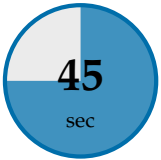
In your team

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THINK & TALK · 1 minute

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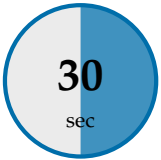
In your team

Write your own letter first — then talk. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

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In your team

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VOTE

Prove from the definition that $\lim_{x \to 6} (x^2 - 9x) = -18$.

The proof begins: *"Let $\varepsilon > 0$. Choose $\delta = \ldots$"* Which δ works for *every* $\varepsilon > 0$?

A. $\delta = \dfrac{\varepsilon}{4}$

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D. $\delta = \min\left\{1, \dfrac{\varepsilon}{3}\right\}$

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C D

VOTE

Prove from the definition that $\lim_{x \to 6} (x^2 - 9x) = -18$.

The proof begins: *"Let* $\varepsilon > 0$. *Choose* $\delta = \ldots$*"* Which δ works for *every* $\varepsilon > 0$?

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A B C D

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A B **C** D

VOTE

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Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C **D**

VOTE

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C. $\delta = \dfrac{\varepsilon}{|x-3|}$

D. $\delta = \min\left\{1, \dfrac{\varepsilon}{3}\right\}$

Hands down. Thank you.

A B C D

ANSWER

Answer

B. $|(x^2 - 9x) - (-18)| = |x - 6|\,|x - 3|$, and the second factor is not a constant: $\delta \leq 1$ keeps x between 5 and 7, where $|x - 3| < 4$; $\delta \leq \varepsilon/4$ does the rest.

A forgets the 1: for $\varepsilon = 44$ it lets in $x = 16$, where $|x^2 - 9x + 18| = 130$.

C lets δ depend on x — but δ is chosen before any x.

D freezes $|x - 3|$ at its value 3 at $x = 6$: for $\varepsilon = 3$ it lets in $x = 6.9$, where $|x^2 - 9x + 18| = 3.51$.

Question 49 — the proof, written out

Scratch work — backwards, from what we want

We want $|(x^2 - 9x) - (-18)| < \varepsilon$. $\quad |x^2 - 9x + 18| = |x - 6|\,|x - 3|$. Tame the factor that is not a constant: if $|x - 6| < 1$, then $5 < x < 7$, so $2 < x - 3 < 4$ and $|x - 3| < 4$; then $|x^2 - 9x + 18| \le 4\,|x - 6| < \varepsilon$ as soon as $|x - 6| < \frac{\varepsilon}{4}$. Both conditions: $\delta = \min\{1, \frac{\varepsilon}{4}\}$.

Question 49 — the proof, written out

Scratch work — backwards, from what we want

We want $|(x^2-9x)-(-18)|<\varepsilon$. $\quad |x^2-9x+18| = |x-6|\,|x-3|$. Tame the factor that is not a constant: if $|x-6|<1$, then $5<x<7$, so $2<x-3<4$ and $|x-3|<4$; then $|x^2-9x+18| \le 4\,|x-6| < \varepsilon$ as soon as $|x-6|<\frac{\varepsilon}{4}$. Both conditions: $\delta = \min\{1, \frac{\varepsilon}{4}\}$.

PROOF Let $\varepsilon > 0$. Choose $\delta = \min\{1, \frac{\varepsilon}{4}\}$; then $\delta > 0$.

Question 49 — the proof, written out

Scratch work — backwards, from what we want

We want $|(x^2-9x)-(-18)|<\varepsilon$. $\quad |x^2-9x+18| = |x-6|\,|x-3|$. Tame the factor that is not a constant: if $|x-6|<1$, then $5<x<7$, so $2<x-3<4$ and $|x-3|<4$; then $|x^2-9x+18| \le 4\,|x-6| < \varepsilon$ as soon as $|x-6|<\frac{\varepsilon}{4}$. Both conditions: $\delta = \min\{1, \frac{\varepsilon}{4}\}$.

PROOF Let $\varepsilon > 0$. Choose $\delta = \min\{1, \frac{\varepsilon}{4}\}$; then $\delta > 0$.

If $0<|x-6|<\delta$, then $|x-6|<1$, so $|x-3|<4$; and $|x-6|<\frac{\varepsilon}{4}$. Hence

$$|(x^2-9x)-(-18)| \;=\; |x-6|\,|x-3| \;\le\; 4\,|x-6| \;<\; 4\cdot\frac{\varepsilon}{4} \;=\; \varepsilon.$$

Question 49 — the proof, written out

Scratch work — backwards, from what we want

We want $|(x^2-9x)-(-18)|<\varepsilon$. $\quad |x^2-9x+18| = |x-6|\,|x-3|$. Tame the factor that is not a constant: if $|x-6|<1$, then $5<x<7$, so $2<x-3<4$ and $|x-3|<4$; then $|x^2-9x+18| \le 4\,|x-6| < \varepsilon$ as soon as $|x-6|<\frac{\varepsilon}{4}$. Both conditions: $\delta = \min\{1, \frac{\varepsilon}{4}\}$.

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So $\lim\limits_{x\to 6}(x^2-9x) = -18$. ∎

Question 49 — the proof, written out

Scratch work — backwards, from what we want

We want $|(x^2-9x)-(-18)|<\varepsilon$. $\quad |x^2-9x+18| = |x-6|\,|x-3|$. Tame the factor that is not a constant: if $|x-6|<1$, then $5<x<7$, so $2<x-3<4$ and $|x-3|<4$; then $|x^2-9x+18| \le 4\,|x-6| < \varepsilon$ as soon as $|x-6|<\frac{\varepsilon}{4}$. Both conditions: $\delta = \min\{1, \frac{\varepsilon}{4}\}$.

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If $0<|x-6|<\delta$, then $|x-6|<1$, so $|x-3|<4$; and $|x-6|<\frac{\varepsilon}{4}$. Hence

$$|(x^2-9x)-(-18)| \;=\; |x-6|\,|x-3| \;\le\; 4\,|x-6| \;<\; 4\cdot\frac{\varepsilon}{4} \;=\; \varepsilon.$$

So $\lim\limits_{x\to 6}(x^2-9x) = -18$. ∎

Four things make it a proof: "let $\varepsilon>0$" first; a δ that depends on ε and on nothing else; the chain, ending in exactly ε; the concluding sentence. A smaller δ works too.

Question 50

$$\lim_{x \to -\infty} \frac{\sqrt{4x^2 + 3}}{5 - 2x} = ?$$

A. -1

B. 2

C. Does not exist: $\frac{\infty}{\infty}$.

D. 1

In your team

Write your own letter first — then talk. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

$$\lim_{x \to -\infty} \frac{\sqrt{4x^2 + 3}}{5 - 2x} = ?$$

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In your team

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THINK & TALK · 1 minute

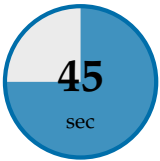
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A. -1

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C. Does not exist: $\frac{\infty}{\infty}$.

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In your team

Write your own letter first — then talk. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

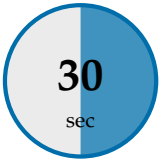
$$\lim_{x \to -\infty} \frac{\sqrt{4x^2 + 3}}{5 - 2x} = ?$$

A. -1

B. 2

C. Does not exist: $\frac{\infty}{\infty}$.

D. 1



In your team

Write your own letter first — then talk. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

$$\lim_{x \to -\infty} \frac{\sqrt{4x^2 + 3}}{5 - 2x} = ?$$

A. -1

B. 2

C. Does not exist: $\frac{\infty}{\infty}$.

D. 1



In your team

Write your own letter first — then talk. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

$$\lim_{x \to -\infty} \frac{\sqrt{4x^2 + 3}}{5 - 2x} = ?$$

A. -1

B. 2

C. Does not exist: $\frac{\infty}{\infty}$.

D. 1



In your team

Write your own letter first — then talk. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

VOTE

$$\lim_{x \to -\infty} \frac{\sqrt{4x^2 + 3}}{5 - 2x} = ?$$

A. -1

B. 2

C. Does not exist: $\frac{\infty}{\infty}$.

D. 1

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C D

VOTE

$$\lim_{x \to -\infty} \frac{\sqrt{4x^2 + 3}}{5 - 2x} = ?$$

A. -1

B. 2

C. Does not exist: $\frac{\infty}{\infty}$.

D. 1

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C D

VOTE

$$\lim_{x \to -\infty} \frac{\sqrt{4x^2 + 3}}{5 - 2x} = ?$$

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Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C D

VOTE

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A. -1

B. 2

C. Does not exist: $\frac{\infty}{\infty}$.

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Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C D

VOTE

$$\lim_{x \to -\infty} \frac{\sqrt{4x^2 + 3}}{5 - 2x} = ?$$

A. -1

B. 2

C. Does not exist: $\frac{\infty}{\infty}$.

D. 1

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C **D**

VOTE

$$\lim_{x \to -\infty} \frac{\sqrt{4x^2 + 3}}{5 - 2x} = ?$$

A. -1

B. 2

C. Does not exist: $\frac{\infty}{\infty}$.

D. 1

Hands down. Thank you.

A B C D

ANSWER

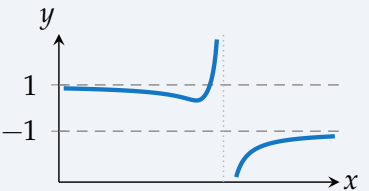
Answer

D. For $x < 0$, $|x| = -x$:
$\sqrt{4x^2+3} = -x\sqrt{4+3/x^2}$ and
$5 - 2x = -x\,(2 - 5/x)$, so
$\lim_{x\to-\infty} \frac{\sqrt{4x^2+3}}{5-2x} = \lim_{x\to-\infty} \frac{\sqrt{4+3/x^2}}{2-5/x} = \frac{2}{2} = 1.$

A is the sign trap ($\sqrt{x^2} = x$ for $x < 0$) — the answer at $+\infty$.

B: $\sqrt{4x^2} = 2|x|$, not $4|x|$.

C: ∞/∞ is an instruction to rewrite.



$y = 1$ on the left, $y = -1$ on the right

STAND UP

"A car is $s(t)$ km from home t hours after noon, and $s'(2) = 60$. Then the car travels exactly 60 km between 2 p.m. and 3 p.m."

Stand up if you think this is TRUE.

Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

STAND UP

"A car is $s(t)$ km from home t hours after noon, and $s'(2) = 60$. Then the car travels exactly 60 km between 2 p.m. and 3 p.m."

Stand up if you think this is TRUE.

Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

Verdict

FALSE. $s'(2) = 60$ km/h is the speed *at the instant* 2 p.m. — the limit of average speeds over shorter and shorter intervals. Over the next hour the car covers 60 km only if it keeps that speed. Example: $s(t) = 15t^2$: $s'(t) = 30t$, $s'(2) = 60$, but $s(3) - s(2) = 135 - 60 = 75$ km. What is true: in a *short* time Δt after 2 p.m. it travels about $60\,\Delta t$ km.

Question 51

$$\lim_{x \to e} \frac{\ln x - 1}{x - e} = ?$$

A. 1

B. 0

C. $\frac{1}{e}$

D. Does not exist — $\frac{0}{0}$.

In your team

Write your own letter first — then talk. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

$$\lim_{x \to e} \frac{\ln x - 1}{x - e} = ?$$

A. 1

B. 0

C. $\frac{1}{e}$

D. Does not exist — $\frac{0}{0}$.

60 sec

In your team

Write your own letter first — then talk. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

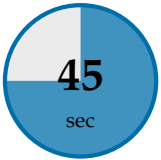
$$\lim_{x \to e} \frac{\ln x - 1}{x - e} = ?$$

A. 1

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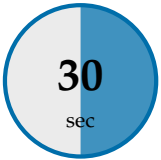
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Question 51

VOTE

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B. 0

C. $\frac{1}{e}$

D. Does not exist — $\frac{0}{0}$.

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C D

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ANSWER

Answer

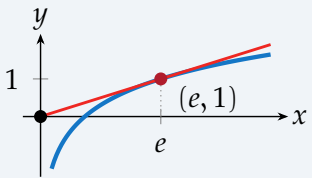
C. $\ln e = 1$, so this is $\lim_{x \to e} \dfrac{\ln x - \ln e}{x - e}$: the derivative of ln at e, in the x-spelling of the definition — $\frac{1}{e}$.

A gives the value $\ln e = 1$ (or the slope at $x = 1$), not the slope at e.

B: the denominator tends to 0 too.

D: every derivative is a $0/0$ limit.

One limit, three names.



tangent at $(e, 1)$: slope $\frac{1}{e}$, through the origin

Question 52

For every $x \neq 1$: $\quad 3-(x-1)^2 \;\leq\; g(x) \;\leq\; 3+2(x-1)^2.$ $\qquad \lim_{x\to 1} g(x) = ?$

A. 3

B. It cannot be found: g itself is not given.

C. It does not exist: g is trapped between two different functions.

D. 3, but only if g is continuous at 1.

In your team

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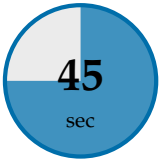
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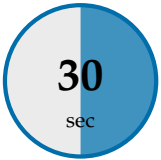
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0
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ANSWER

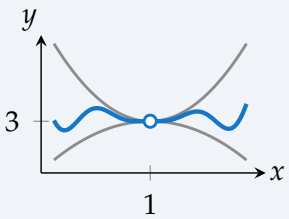
Answer

A. $\lim_{x\to 1}\big(3-(x-1)^2\big)=3$ and $\lim_{x\to 1}\big(3+2(x-1)^2\big)=3$, so by the Squeeze Theorem $\lim_{x\to 1} g(x)=3$.

B: a limit needs no formula — the bounds are enough.

C: different bounds (at $x=2$: 2 and 5), the same limit.

D: near 1, never at 1 — $g(1)$ may be anything.



squeezed at $(1,3)$

By heart (1) — the definitions

The limit

$\lim_{x \to a} f(x) = L$: for every $\varepsilon > 0$ there is a $\delta > 0$ such that
$0 < |x - a| < \delta \;\Rightarrow\; |f(x) - L| < \varepsilon$.
ε first, then δ; near a, not at a.

Continuity at a

$f(a)$ exists; $\lim_{x \to a} f(x)$ exists; they are equal. Three common ways to fail: removable, jump, infinite.

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The Intermediate Value Theorem

f continuous on $[a, b]$, N strictly between $f(a)$ and $f(b)$ $\Rightarrow$ $f(c) = N$ for some c in (a, b).

The derivative at a point

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h} = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$$

the tangent slope, the instantaneous rate, the velocity: one limit, three names.

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The habits

Near, not at. $0/0$, ∞/∞, $\infty - \infty$ *and* $0 \cdot \infty$ *are instructions. Check the hypotheses before you use a theorem.* lim *on every line until the limit is taken. A rate has units.*

By heart (2) — the techniques

When substitution gives $0/0$

Factor and cancel; the conjugate; combine the fractions; an absolute value — go one-sided; squeeze.

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Far away, and asymptotes

Divide by the highest power; $\sqrt{x^2} = |x|$, which is $-x$ for $x < 0$.

Asymptotes: horizontal $y = L$ when $\lim_{x\to\infty} f(x) = L$ or $\lim_{x\to-\infty} f(x) = L$ (the graph may cross it); vertical $x = a$ when a one-sided limit at a is $\pm\infty$.

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A piecewise function

Continuous at the seam when both one-sided limits equal the value there.

STAND UP

"If f is differentiable at a, then $\lim_{h\to 0}\dfrac{f(a+2h)-f(a)}{h}=f'(a)$."

Stand up if you think this is TRUE.

Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

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Verdict

FALSE — it is $2f'(a)$. Put $k=2h$; as h tends to 0, so does k:

$$\lim_{h\to 0}\frac{f(a+2h)-f(a)}{h}=\lim_{h\to 0}2\cdot\frac{f(a+2h)-f(a)}{2h}=2\lim_{k\to 0}\frac{f(a+k)-f(a)}{k}=2f'(a).$$

Example: $f(x)=x$: the quotient is $\frac{2h}{h}=2$, while $f'(a)=1$. The definition asks for the same h in the numerator and in the denominator.

Tonight — three things

1. Without notes, one exercise of each kind that you once got wrong: a limit by factoring; a one-sided limit with an absolute value; an ε–δ proof with the min trick; a continuity check with the three conditions; a derivative from the definition, with lim on every line.

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Next:

the derivative as a function, the rules, inverse, exponential and logarithmic functions — in questions.

EXIT TICKET

Tear a quarter page from a notebook — one per team.

1. Team name; names of those present today.
2. One sentence: *the one kind of question about limits, continuity or the derivative at a point that I still cannot do is … (a kind of question, not a chapter).*
3. Your muddiest point — the one thing still unclear.
4. One moment today that surprised you.

Give it to the professor or the course assistant as you go. Thank you.